PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Diagnostic accuracy of ultrasound, computed tomography, and their combination in detecting cervical lymph node metastasis in patients with papillary thyroid cancer: a systematic review and meta-analysis
AUTHORS	Yang, Jian; Zhang, Fengyan; Qiao, Ying

VERSION 1 – REVIEW

REVIEWER	Mollah, Md Nurul Haque University of Rajshahi, Statistics
REVIEW RETURNED	15-Jun-2021

GENERAL COMMENTS	Review Report Paper Title: Cervical lymph node metastasis in papillary thyroid cancer: a meta-analysis of ultrasound and CT for diagnosis and management Journal Name: BMJ open Manuscript ID: bmjopen-2021-051568 General Comments: In this manuscript, the authors performed the meta-analysis to investigate the prediction performance of cervical lymph node metastasis (CLNM) with papillary thyroid cancer (PTC) patients by the ultrasound (US), computed tomography (CT) and combination of US and CT. The performance was measured using sensitivity, specificity, odds ratio (OR), positive likelihood ratio (LR+) and negative likelihood ratio (LR-). Finally, they suggested the combined method instead of the individual US and CT for CLMA prediction based on their computational results. The combined approach was selected by giving the emphasis on larger sensitivity instead of specificity and smaller odd ratio (OR). Therefore, the paper seems interesting. However, in my opinion, it needs major revision before going to the publication according to the following instructions. Specific Comments: 1. The author should create a section by introducing the US, CT and their combined approaches about how they predict CLMN. 2. They should define their used performance measures sensitivity, specificity, odds ratio (OR), positive likelihood ratio (LR+) and negative likelihood ratio (LR-) with respect to their research problems. 3. They should explain why they gave emphasis on larger sensitivity instead of specificity and smaller odd ratio (OR) to

	select the better approach. They should also consider the LT+ and LR- results to select the better method. 4. They performed model heterogeneity test to select one of fixed or random effect models. But model heterogeneity test cannot confirm the adequacy or goodness of fit of the model, which may provide misleading results (Chen et al, 2015,). Therefore, my suggestion is to compare your results with the goodness of fit test results for your models. 5. There are several typo errors in throughout the manuscript, for example, 'Foreplot' should be 'Forest plot' in lines 28-29 page-9, space problem when they used mathematical notation in lines 16-20, and so on. 6. The combination of US and CT increased sensitivity by 30-50% on patient-based analysis. Is it correct? Explain how? Reference: Chen Z, Zhang G, Li J. Goodness-of-fit test for meta-analysis. Sci Rep. 2015 Nov 23; 5:16983. https:// doi.org/10.1038/srep16983 PMID: 26592212; PMCID: PMC4655309.

REVIEWER	Pardal-Refoyo , José Luis IBSAL, OTOLARYNGOLOGY
REVIEW RETURNED	16-Oct-2021

GENERAL COMMENTS	is a systematic review and meta-analysis to evaluate the sensitivity, specificity and odds ratio of ultrasound, computerized axial tomography and both for the diagnosis of nodal metastases of papillary thyroid carcinoma. The article is correct both in its methodology and in the presentation of its results.

REVIEWER	Karunakaran, Poongkodi Government Mohan Kumaramangalam Medical College, Endocrine surgery
REVIEW RETURNED	22-Oct-2021

GENERAL COMMENTS	The authors have done extensive analysis on the utility of US and CT in detecting cervical lymph-node mets in papillary thyroid cancer. However, there are many concerning issues in the which the authors need to address. What do the authors mean by level-by-level basis or patient based analysis. What do the authors mean by neck level, central level or lateral neck level. Kindly define the boundaries of each compartment. Similarly, what do the authors mean by patient level, central patient level, and lateral patient level. Language should be improved. Grammar and syntax error has to be addressed. The manuscript needs major revision.

VERSION 1 – AUTHOR RESPONSE

Reviewer #1

1. The author should create a section by introducing the US, CT and their combined approaches about how they predict CLMN.

Response: We thank the Reviewer. We added some introduction about them. Please see the Introduction (2nd paragraph).

2. They should define their used performance measures sensitivity, specificity, odds ratio (OR), positive likelihood ratio (LR+) and negative likelihood ratio (LR-) with respect to their research problems.

Response: We thank the Reviewer. Sensitivity, specificity, DOR, LR+, and LR- were calculated for the detection of positive CLNMs by US and/or CT at the neck compartment level or patient level. It was clarified at the end of the Introduction.

3. They should explain why they gave emphasis on larger sensitivity instead of specificity and smaller odd ratio (OR) to select the better approach. They should also consider the LT+ and LR- results to select the better method.

Response: We thank the Reviewer. We gave emphasis on sensitivity because the aim of CLNM screening is to detect and sample all suspicious lymph nodes to avoid letting positive lymph nodes in place, which would affect the prognosis of the patient. Specificity is given less emphasis because false positives can increase the morbidity of the dissection but at least will not affect the oncological prognosis.

4. They performed model heterogeneity test to select one of fixed or random effect models. But model heterogeneity test cannot confirm the adequacy or goodness of fit of the model, which may provide misleading results (Chen et al, 2015,). Therefore, my suggestion is to compare your results with the goodness of fit test results for your models.

Response: We thank the Reviewer for the comment. We agree that the analysis of heterogeneity is controversial. Still, analyzing heterogeneity and selecting the model accordingly is an approach advocated by the Cochrane Collaboration[6].

5. There are several typo errors in throughout the manuscript, for example, 'Foreplot' should be 'Forest plot' in lines 28-29 page-9, space problem when they used mathematical notation in lines 16-20, and so on.

Response: The manuscript was proofread.

6. The combination of US and CT increased sensitivity by 30-50% on patient-based analysis. Is it correct? Explain how?

Response: We thank the Reviewer. It is an error that occurred during manuscript preparation in translation. It was revised as "The combination of US and CT increased the sensitivity from 41%-49% for the individual modalities to 64% for combined modalities in the patient-based analysis."

Reviewer #2

is a systematic review and meta-analysis to evaluate the sensitivity, specificity and odds ratio of ultrasound, computerized axial tomography and both for the diagnosis of nodal metastases of papillary thyroid carcinoma.

The article is correct both in its methodology and in the presentation of its results.

Response: We thank the Reviewer for taking the time to review our manuscript and for the comments.

Reviewer #3

What do the authors mean by level-by-level basis or patient based analysis.

Response: We thank the Editor. In the level-by-level (now revised as “neck level-based”), the results were analyzed in terms of positive lymph nodes in each neck level, independently. In the patient-based levels, positive lymph nodes were considered irrespective of the neck level. It was clarified in the manuscript.

What do the authors mean by neck level, central level or lateral neck level. Kindly define the boundaries of each compartment.

Response: We thank the Reviewer. The central neck refers to Groups VI and VII lymph nodes. The lateral neck refers to Groups I to V lymph nodes. They were clarified in the Introduction.

Similarly, what do the authors mean by patient level, central patient level, and lateral patient level.

Response: We thank the Reviewer. The patient level meant the whole patient. Central and lateral levels mean central and lateral metastasis.

Language should be improved. Grammar and syntax error has to be addressed.

Response: The manuscript was proofread.

References

1. Suh CH, Baek JH, Choi YJ, et al. Performance of CT in the Preoperative Diagnosis of Cervical Lymph Node Metastasis in Patients with Papillary Thyroid Cancer: A Systematic Review and Meta-Analysis. *AJNR Am J Neuroradiol* 2017;38(1):154-61. doi: 10.3174/ajnr.A4967 [published Online First: 2016/10/30]
2. Cho SJ, Suh CH, Baek JH, et al. Diagnostic performance of CT in detection of metastatic cervical lymph nodes in patients with thyroid cancer: a systematic review and meta-analysis. *Eur Radiol* 2019;29(9):4635-47. doi: 10.1007/s00330-019-06036-8 [published Online First: 2019/02/27]
3. Raijmakers PG, Paul MA, Lips P. Sentinel node detection in patients with thyroid carcinoma: a meta-analysis. *World J Surg* 2008;32(9):1961-7. doi: 10.1007/s00268-008-9657-y [published Online First: 2008/07/03]
4. Wu LM, Gu HY, Qu XH, et al. The accuracy of ultrasonography in the preoperative diagnosis of cervical lymph node metastasis in patients with papillary thyroid carcinoma: A meta-analysis. *Eur J Radiol* 2012;81(8):1798-805. doi: 10.1016/j.ejrad.2011.04.028 [published Online First: 2011/05/04]
5. Zhao H, Li H. Meta-analysis of ultrasound for cervical lymph nodes in papillary thyroid cancer: Diagnosis of central and lateral compartment nodal metastases. *Eur J Radiol* 2019;112:14-21. doi: 10.1016/j.ejrad.2019.01.006 [published Online First: 2019/02/20]
6. Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for Systematic Reviews of Interventions version 6.1. London: Cochrane Collaboration 2020.

VERSION 2 – REVIEW

REVIEWER	Karunakaran, Poongkodi Government Mohan Kumaramangalam Medical College, Endocrine surgery
REVIEW RETURNED	24-Jan-2022

GENERAL COMMENTS	I appreciate the authors for the revision of the manuscript as per the reviewer's/ editor's suggestion. It has improved the quality and readability of the manuscript. I have the following concerns which need to be addressed. 1. In the table 2. The authors have mentioned that the results are expressed as n (95% CI; I2, %). However, the results are expressed as 0.41(0.36-0.4,95.5) which appears to be different format. The authors should mention the format clearly and rectify any mistakes in the entire table. 2. The authors emphasize on the sensitivity but have presented sROCs for specify in both the neck based and patient-based analysis. Moreover, they have to derive the AUC (area under the curve) for each modality which be a better parameter to define the diagnostic accuracy. 3. Importantly, the authors should mention p-value to compare the different modality US vs Ct, US vs combined modality or CT vs combined modality. 4. The authors should correct the fonts, spacing in the reference section as per the recommended reference style.

VERSION 2 – AUTHOR RESPONSE

Responses to reviewers' comments:

Reviewer: 3

Dr. Poongkodi Karunakaran, Government Mohan Kumaramangalam Medical College, Madras Medical College

Comments to the Author:

I appreciate the authors for the revision of the manuscript as per the reviewer's/ editor's suggestion. It has improved the quality and readability of the manuscript.

Response: We also thank the reviewer for taking the time to check our revised article and the encouraging comments.

I have the following concerns which need to be addressed.

1. In the table 2. The authors have mentioned that the results are expressed as n (95% CI; I2, %). However, the results are expressed as 0.41(0.36-0.46,95.5) which appears to be different format. The authors should mention the format clearly and rectify any mistakes in the entire table.

Response: Thanks for this suggestion! The data 0.41(0.36-0.46,95.5) represents n=0.41, 95%CI (0.36-0.46), I2=95.5%. We have added the heterogeneity statistics (I^2%) in the revised Table 2. However, due to too much data, the arrangement of Table 2 will be affected. Therefore, it is kept as is and modify to n (95% CI, I2 %). If not clear, please contact me.

2. The authors emphasize on the sensitivity but have presented sROCs for specify in both the neck based and patient-based analysis. Moreover, they have to derive the AUC (area under the curve) for each modality which be a better parameter to define the diagnostic accuracy.

Response: We thank the reviewer for this great suggestion. The included studies presented sROCs for specify in both the neck based and patient-based analysis, so we pooled the corresponding data for this meta-analysis. Besides, only two included studies analyzed AUC (Eun et al. 2018 and Yoon et al. 2011). However, the study by Eun et al. compared CT vs US vs fine needle aspiration cytology (FNAC) in terms of AUC. The study by Yoon et al compared CT vs CT/US combined modality. The data from these two studies could not be combined for a meta-analysis.

3. Importantly, the authors should mention p-value to compare the different modality US vs Ct, US vs combined modality or CT vs combined modality.

Response: We used Meta-Disc software (version 1.4, Madrid, Spain) for performing this meta-analysis. However, due to some inherent statistical shortcomings of separate pooling of sensitivities and specificities and the Littenberg and Moses model, between-study variance could not be calculated.

4. The authors should correct the fonts, spacing in the reference section as per the recommended reference style.

Response: As the reviewer suggested, we have corrected the fonts, spacing in the reference section according to the recommended reference style of your journal. Please check the re-submitted manuscript.